

ES VIDEO LEZIONI 2022

LE 34

1) - Si lancia una moneta n volte. Determinare la probabilità di ottenere esattamente z teste.

Svolgimento

X : n° delle T in n lanci
evento che ci interessa $\{X=z\}$ e cioè $P(X=z)$

Spettro: $S_X = \{0, 1, 2, \dots, n-1, n\}$

e la $P(X=z) > 0 \forall z$
appartiene allo spettro

X ha legge B con parametri n e p (Prob teste)

$X \sim B(n, p)$ $p = P(T)$

legge binomiale
3 fattori

$$P(X=z) = \underbrace{\binom{n}{z}}_{1^\circ} \cdot \underbrace{p^z}_{2^\circ} \underbrace{(1-p)^{n-z}}_{3^\circ}$$

1°) n° modi x cui esce teste $\binom{n}{z}$

2°) n lanci z f. $p = p^z$

3°) insuccessi $(1-p)^{n-z}$

- Si lancia un dado n° volte e indichiamo con N il "punteggio"

N è discreto non più come prima

X è il numero delle teste T in N lanci

$P(X=z)$

Uniamo le alternative $\Rightarrow$ non so quante volte dico lanciare ~~oblio~~ e ho legge binomiale

$\{N=1\}, \{N=2\}, \dots, \{N=5\}, \{N=6\}$

$$P(X=z) = \sum_{n=1}^6 P(X=z | N=n) \cdot P(N=n) = \frac{1}{6} \sum_{n=2}^6 P(X=z | N=n) =$$

$$= \frac{1}{6} \sum_{n=2}^6 \binom{n}{z} p^{n-z}$$

3) X d.l., $F_X(x, a)$ $a > 0$

$$F_X(x, a) = \begin{cases} 0 & x \leq 0 \\ x^a & 0 < x < 1 \\ 1 & x \geq 1 \end{cases}$$

Var. d. di tipo continuo, x ha
e' intervallo

$$f_X(x) = \begin{cases} a x^{a-1} & 0 < x < 1 \\ 0 & \text{altimenti} \end{cases}$$

Calcolare: $k \in \mathbb{N}$, $\mu'_k := E(X^k)$

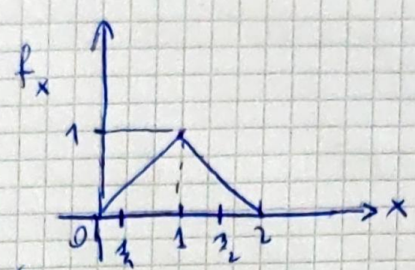
Soluz:

$$\mu'_k = E(X^k) = \int_0^1 x^k f_X(x, a) dx = \int_0^1 a x^k \cdot x^{a-1} dx =$$

$$= a \int_0^1 x^{k+a-1} dx = a \left. \frac{x^{k+a}}{k+a} \right|_0^1 = \frac{a}{k+a}$$

4) $f_X(x) = \begin{cases} x & 0 < x < 1 \\ 2-x & 1 \leq x < 2 \\ 0 & \text{altre} \end{cases}$

$S_X = (0, 2)$



Determinare il reale ξ per il quale risulta $P(\frac{1}{4} < X \leq \xi) = P(\xi < X \leq \frac{3}{2})$

Soluz: $\boxed{\xi = 1}$

$P(\frac{1}{4} < X \leq \xi) = \int_{\frac{1}{4}}^{\xi} x dx = \frac{x^2}{2} \Big|_{\frac{1}{4}}^{\xi} = \frac{1}{2} (\xi^2 - \frac{1}{16})$ Fori: calcoli

$P(\xi < X \leq \frac{3}{2}) = \int_{\xi}^{\frac{3}{2}} f_X(x) dx = \int_{\xi}^1 x dx + \int_1^{\frac{3}{2}} (2-x) dx$

5) $Y \sim N(1, \sigma^2)$ tale che $P(-1 < Y \leq 3) = 0,754$

Guassiana

Determinare $\sigma^2 = D^2(Y)$ Varianza

Soluz.

$0,754 = P(-1 < Y \leq 3) = P(-\frac{2}{\sigma} < Z \leq \frac{2}{\sigma}) = 2P(0 \leq Z \leq \frac{2}{\sigma})$

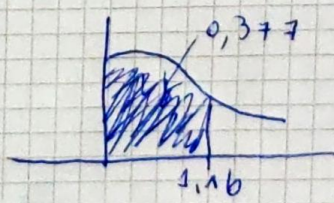
Tabella distribuzione gaussiana standard $\frac{0,754}{2}$

$P(0 \leq Z \leq \frac{2}{\sigma}) = 0,377$

$\frac{2}{\sigma} = z \Rightarrow \sigma = \frac{2}{z}$

$\int_0^z \phi(t) dt = 0,377$

$z = 1,16$



$\frac{2}{\sigma} = 1,16 \Rightarrow \sigma = \frac{2}{1,16} \Rightarrow \sigma^2 = \left(\frac{2}{1,16}\right)^2$

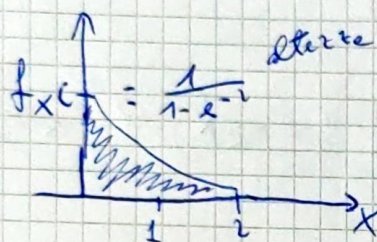
0) Es Standard

L: Coto di un quadrato

L^2 d. assolutamente continuo con funz. dens. prob. f. d. p) segue da

$$f_L(x) = \begin{cases} c \cdot e^{-x} & 0 \leq x \leq 2 \\ 0 & \text{altrimenti} \end{cases}$$

Determinare c



$$1 = \int_0^2 c \cdot e^{-x} dx = c \cdot \int_0^2 e^{-x} dx \Rightarrow 1 = c \cdot e^{-x} \Big|_0^2$$
$$\Rightarrow 1 = c(1 - e^{-2}) \Rightarrow c = \frac{1}{1 - e^{-2}}$$

$$f_L(x) = \begin{cases} \frac{e^{-x}}{1 - e^{-2}} & 0 \leq x \leq 2 \\ 0 & \text{altrimenti} \end{cases}$$

Determinare la Probabilità primitiva del quadrato $0 \leq x \leq 2$

$$\text{evento: } \{L < \frac{1}{2}\} \Rightarrow \{L < \frac{1}{2}\}$$

$$P(L < \frac{1}{2}) = \int_0^{\frac{1}{2}} \frac{e^{-x}}{1 - e^{-2}} dx = \dots = \frac{1 - e^{-1/2}}{1 - e^{-2}}$$

Proprietà Z RV distribuita gli $P \in \mathcal{B}$.

1) $f_x(x) \geq 0$

2) $\int_{\mathbb{R}} f_x(x) dx = 1$

a) $F_x(x)$ non decrescente

b) $x \in \mathbb{R}, \lim_{\varepsilon \rightarrow 0^+} F_x(x+\varepsilon) = F_x(x)$

c) $\lim_{x \rightarrow -\infty} F_x(x) = 0$

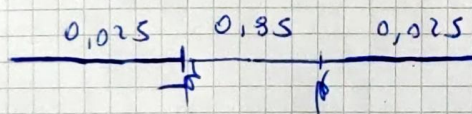
$\lim_{x \rightarrow +\infty} F_x(x) = 1$

Altro esercizio su Gaussiana e Tabella

7) $Z \sim N(0, 1)$

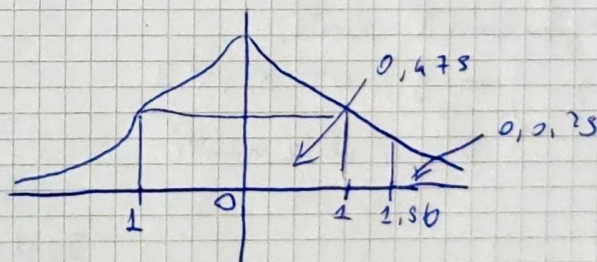
Si determini $\beta \in \mathbb{R}^+$. Probabilità che Z assume un valore esterno all'intervallo $(-\beta, \beta)$ vale 0.05. $\leftarrow$ x.s. simmetrica

Quindi $P(|Z| \leq \beta) = 0.95 \Leftrightarrow 2P(Z \leq \beta) = 0.95 \Leftrightarrow$



$\Leftrightarrow P(Z \leq \beta) = 0.475$

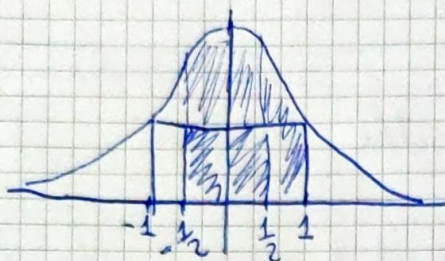
$\beta = 1.96$



8) $X \sim N(1, 4)$

1) $P(0 \leq X \leq 3) = P\left(\frac{0-1}{2} \leq Z \leq \frac{3-1}{2}\right) = P\left(-\frac{1}{2} \leq Z \leq \frac{1}{2}\right) = P\left(-\frac{1}{2} \leq Z \leq 0\right) + P\left(0 \leq Z \leq \frac{1}{2}\right) = P\left(0 \leq Z \leq \frac{1}{2}\right) + P\left(0 \leq Z \leq \frac{1}{2}\right) \approx 0,19 + 0,34 \approx 0,53$

Negative mean zero probability



Importante

2) $P(X=1 | 0 \leq X \leq 3) =$

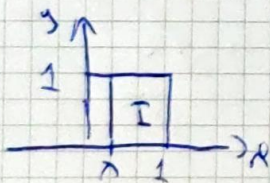
$\{X=1\} \cap \{0 \leq X \leq 3\} = \{X=1\}$

$= \frac{P(X=1)}{P(0 \leq X \leq 3)} = 0$

$P(X=1) = 0$ since it's continuous

3) (X, Y) jointly distributed.

$f(x, y) = \begin{cases} x+y, & 0 \leq x, y \leq 1 \\ 0, & \text{otherwise} \end{cases}$



It's a square

a) Show that X and Y are independent: (Note: we need to show that $f(x, y) = f_X(x) \cdot f_Y(y)$)

$f_X(x) = \int_0^1 f(x, y) dy = \int_0^1 (x+y) dy = x \cdot y + \frac{y^2}{2} \Big|_0^1 = x + \frac{1}{2}$

$f_Y(y) = \int_0^1 f(x, y) dx = \int_0^1 (x+y) dx = \frac{x^2}{2} + xy \Big|_0^1 = \frac{1}{2} + y$

$f(x, y) = (x + \frac{1}{2})(\frac{1}{2} + y) = \frac{1}{4} + \frac{xy}{2} + \frac{x}{2} + \frac{y}{2}$

$y \in (0, 1)$

$$f_y(y) = y + 1/2$$

$$x+y \neq (x+\frac{1}{2})(y+\frac{1}{2}) \text{ Non zero independent}$$

b) valore a pro correlato

è utile bisogna trovare la variante, e in ^{di più} ~~già~~ 0,01 per

$$\rho_{xy}(x, y) = \frac{E[(x - \mu_x)(y - \mu_y)]}{\sigma_x \sigma_y} = \frac{E(xy) - \mu_x \mu_y}{\sigma_x \sigma_y}$$

$$\mu_x = E(x); \quad \mu_y = E(y)$$

$$f_X(x) = \begin{cases} x + \frac{1}{2}, & 0 \leq x \leq 1 \\ 0, & \text{alt.} \end{cases}$$

$$f_y(y) = \begin{cases} y + \frac{1}{2} & 0 \leq y \leq 1 \\ 0 & \text{altre} \end{cases}$$

risulta che X e Y sono omologanti: $\Rightarrow \mu_x = \mu_y$

Glucosio M_x

$$M_x = E(X) = \int_0^1 x \left(x + \frac{1}{2}\right) dx = \int_0^1 x^2 dx + \frac{1}{2} \int_0^1 x dx = \frac{x^3}{3} \Big|_0^1 + \frac{1}{2} \frac{x^2}{2} \Big|_0^1 = \frac{1}{3} + \frac{1}{4} = \frac{7}{12}$$

$$\sigma_{E(X \cdot Y)}$$

$$F[g(\underline{x})] = \int_{\mathbb{R}} g(\underline{x}) f_{\underline{x}}(\underline{x}) d\underline{x}$$

elie immutabile

$$E(X \cdot Y) = \int_I \int_I xy(x+y) dx dy = \int_I \int_I x^2 y dx dy + \int_I \int_I xy^2 dx dy = \text{symmetrisch}$$

$$+ \int_0^1 dx \cdot \int_0^1 dy \cdot xy^2 = \int_0^1 x dx \int_0^1 y^2 dy = \int_0^1 x dx \cdot \frac{y^3}{3} \Big|_0^1 = \frac{1}{3} \cdot \frac{x^2}{2} \Big|_0^1 = \frac{1}{6}$$

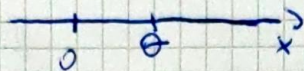
$$* \text{Contorno} = \frac{1}{3} \quad \text{aka} \quad \frac{1}{6} + \frac{1}{6} = \frac{1}{3}$$

Quindi $E(\cancel{X}Y) \neq M_x M_y = \frac{1}{3} - \frac{49}{144} = -\frac{1}{144} \neq 0$

Per ± 0 due carichi, e lo sono in negativo, se uno è o l'altro è maggiore ciascuno e c'è una tendenza negativa

LE 7 35

1) $\underline{X} = (x_1, x_2, \dots, x_m)$, x e.e.v. $f_x(x, \theta) = \begin{cases} e^{-(x-\theta)}, & x > \theta \\ 0, & \text{altz} \end{cases}$ $\theta > 0, \textcircled{H} \rightarrow (0, \infty)$



Belimierung

$$\int_0^{\infty} e^{-(x-\theta)} dx = \int_0^{\infty} e^{-(x-\theta)} d(x-\theta) = e^{-(x-\theta)} \Big|_0^{\infty} = 1 - 0 = 1$$

POP überprüfen

2) $X \in \mathcal{M}_2$ discreto. In $S_X = \{-N, -N+1, \dots, 0, 1, \dots, 2N-1, 2N\}$

Determinare P.B. : $\textcircled{P}$ $k \in S_X, \sum_{k \in S_X} P(X=k) = e \cdot 2^{-|k|}$ in $c > 0$

$$1 = c \sum_{j=-N}^{2N} 2^{-|j|} = c \left[\sum_{j=-N}^{-1} 2^j + \sum_{j=0}^{2N} 2^{-j} \right] \text{ per moltiplicare } j: -i$$

$$= c \left[\sum_{i=1}^N 2^{-i} + \sum_{j=0}^{2N} 2^{-j} \right] = c \left[\sum_{i=1}^N \left(\frac{1}{2}\right)^i + \sum_{j=0}^{2N} \left(\frac{1}{2}\right)^j \right] = 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} = \frac{7}{8}$$

$$c = \frac{8}{7}$$

Determinare la legge di $Y = X^2$

$$S_X = \{-4, -3, -2, -1, 0, 1, 2, 3, 4\}$$

$$S_Y = \{0, 1, 4, 9, 16\}$$

$$P(Y=16) = P(X=4) = c \cdot 2^{-4}$$

$$P(Y=9) = P(X=3) = c \cdot 2^{-3}$$

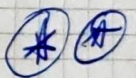
$$P(Y=4) = P(X=-2) + P(X=2) = c \cdot 2^{-2} + c \cdot 2^{-2}$$

$$P(Y=0) = P(X=0) = c \cdot 1 = c$$

$$P(Y=1) = P(X=-1) + P(X=1) = c \cdot 2^{-1} + c \cdot 2^{-1}$$

Se $N=4$

$$S_Y = \{0, 1, 4, 9, 16\}$$

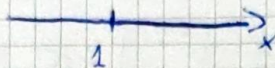


IMPORTANT

3) $\alpha > 0$ dato

⊛ IMP.

$$F_X(x) = \begin{cases} 1 - \frac{1}{x^\alpha}, & x > 1 \\ 0, & \text{altrimenti} \end{cases}$$



$k \in \mathbb{N}$

Def. f.d.p. Formula di Moore $M_k = E(X^k) = \int_{-\infty}^{+\infty} f_X(x) dx$

$$f_X(x) = \begin{cases} \alpha x^{-\alpha-1}, & x > 1 \\ 0, & \text{altrimenti} \end{cases} = \begin{cases} \alpha \cdot \frac{1}{x^{\alpha+1}}, & x > 1 \\ 0, & \text{altrimenti} \end{cases} \quad \left\{ \begin{array}{l} \text{Per } \alpha=2 \end{array} \right.$$

$$M_k = E(X^k) = \int_{-\infty}^{+\infty} x^k f_X(x) dx = \int_1^{+\infty} \frac{\alpha}{x^{\alpha+1}} x^k dx = \alpha \int_1^{+\infty} x^{-(\alpha+1)+k} dx =$$

$$= \alpha \left[\frac{x^{-(\alpha+1)+k+1}}{-(\alpha+1)+k+1} \right]_1^{+\infty} = \alpha \left[\frac{x^{k-\alpha}}{k-\alpha} \right]_1^{+\infty} = \begin{cases} \frac{\alpha}{k-\alpha} & k > \alpha \\ \infty & k \leq \alpha \end{cases}$$

Se $k = \alpha$

$$\alpha \int_1^{+\infty} x^{-\alpha-1+\alpha} dx$$

Quando $k \leq \alpha$ esiste M_k Metodo dei momenti

LEZ 36

1) X o.c. f.d.p.

distribuzione $\bar{x}$ di LAPLACE (caso)

$$f_x(x; \mu, \sigma) = \frac{1}{\sigma} e^{-\frac{|x-\mu|}{\sigma}}$$

 $x \in \mathbb{R}$ $\mu \in \mathbb{R}$

fattore di localizzazione

 $\sigma > 0$

fattore di scala

Alcuni risultati:

$$E(X) = \mu$$



$$\Rightarrow \mu_1' = \mu$$

$$D^2(X) = \sigma^2$$

$$\Rightarrow \mu_2' - (\mu_1')^2 = \sigma^2$$

$$\Rightarrow \mu_2' = (\mu_1')^2 + \sigma^2$$

$$\mu = 2, \quad \sigma_1 = 1, \quad \sigma_2 = 2$$

$$\mu = \mu_1'$$

$$\sigma^2 = \mu_2' - (\mu_1')^2 \Rightarrow \sigma^2 = \frac{1}{2} [\mu_2' - (\mu_1')^2] \Rightarrow \sigma = \sqrt{\frac{\mu_2' - (\mu_1')^2}{2}}$$

$$\hat{A}_{MM} = \bar{X}, \quad \hat{B}_{MM} = \sqrt{\frac{\bar{X}^{(2)} - (\bar{X})^2}{2}}$$

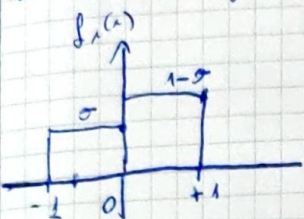
7) legge di probabilità

X m.o. m. continuo

$\theta \in (0,1)$

$$f(x) = \begin{cases} \theta & , -1 \leq x \leq 0 \\ 1-\theta & , 0 \leq x \leq 1 \\ 0 & , \text{altrimenti} \end{cases}$$

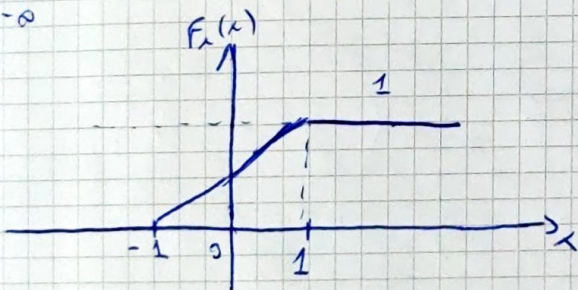
Determinare e graficare della f.d. di X .



$$F_X(x) = \begin{cases} 0 & , x \leq -1 \\ \theta(x+1) & , -1 \leq x \leq 0 \\ \theta + (1-\theta)x & , 0 \leq x \leq 1 \\ 1 & , x \geq 1 \end{cases}$$

$$\int_{-\infty}^x f_X(x) dx = \int_{-1}^x \theta dx = \theta(x+1)$$

$$\int_{-\infty}^x f_X(x) dx = 0 + \theta + \int_0^x (1-\theta) dx = \theta + (1-\theta)x$$



Calore

Determiniamo la mediana

$$F_X(M_e) = \frac{1}{2} \quad \text{e} \quad \theta = \frac{1}{2}$$

$$\theta + (1-\theta)M_e = \frac{1}{2} \Rightarrow M_e = \frac{\frac{1}{2} - \theta}{1-\theta} = \frac{1-2\theta}{2(1-\theta)}$$

$$\text{If } \theta = \frac{1}{2}$$

$$\frac{4 - \frac{3}{2}}{2 \cdot \frac{3}{2}} = \frac{1}{3}$$

$$\text{If } \theta > \frac{1}{2}$$

$$O(\text{He}+1) = \frac{1}{2}$$

$$\text{He}+1 = \frac{1}{2\theta}$$

$$\text{He} = \frac{1}{2\theta} - 1 = \frac{1-\theta}{2\theta}$$

- Déterminer la Médie $E(x)$

$$\text{He} = \begin{cases} \frac{1-\theta}{2\theta}, & \theta > \frac{1}{2} \\ \frac{1-\theta}{2(1-\theta)}, & \theta \leq \frac{1}{2} \end{cases}$$

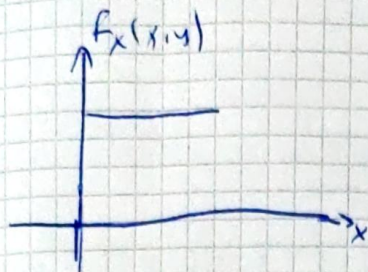
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$$\begin{aligned} E(x) &= \int_{-1}^1 x f_x(x) dx = \int_{-1}^0 x \cdot \theta dx + \int_0^1 x(1-\theta) dx = \theta \frac{x^2}{2} \Big|_{-1}^0 + (1-\theta) \frac{x^2}{2} \Big|_0^1 = \\ &= \frac{1}{2} [\theta + (1-\theta)] = \frac{1-\theta}{2} = \frac{1}{2} - \theta \end{aligned}$$

So $\theta = \frac{1}{2}$ when

$$3) F_x(x, a) = \begin{cases} \frac{1}{1-ax}, & x < 0 \\ 1, & x \geq 0 \end{cases}$$

Per quali valori di a F_x è una f.d.

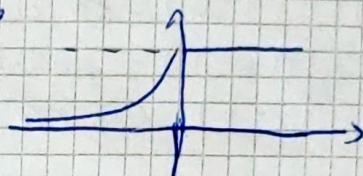


$$F'(x, a) = \frac{0 - (a)}{(1-ax)^2} = \frac{a}{(1-ax)^2}$$

Se $a > 0$ la funzione è crescente quindi $a > 0$

Se $a = 0$, rimane 1 e non è una f.d. $F(x) = \frac{a}{(1-ax)^2} \Rightarrow a > 0$

$$\lim_{x \rightarrow 0^-} \frac{1}{1-ax} = 1 \quad \text{lim sinistro}$$



Aggiungiamo una v.e. $Y = \frac{1}{X}$, Dct. f.d. di Y .

$$F_Y(y) = P(Y \leq y) = P\left(\frac{1}{X} \leq y\right) = P\left(X \leq \frac{1}{y}\right) = P\left(\frac{1}{y} \leq X \leq 0\right) = x^0$$

$$= F_X(0) - F_X\left(\frac{1}{y}\right) = 1 - \frac{1}{1-\frac{a}{y}} = \frac{a}{a-y}$$

$$F_Y(y) = \begin{cases} \frac{a}{a-y}, & y \geq 0 \quad 1^o \\ 1, & y < 0 \quad 2^o \end{cases}$$

$$\text{Per } y \geq 0 \quad 2^o$$

$$F_Y(y) = P(Y \leq y) = P\left(\frac{1}{X} \leq y\right) =$$

$$= P\left(X \geq \frac{1}{y}\right) = P(X \leq 0) =$$

$$= 0 + F_X(0) = 1$$

Per $a = 1$, $F_X(x, 1) = F_Y(y, 1)$ $y = 1/x$

$$F_X(x, 1) = F_Y\left(\frac{1}{x}, 1\right)$$

$$x \stackrel{a=1}{\sim} y$$

$$\frac{1}{1-x} \quad \text{e} \quad \frac{1}{1-y}$$

$$\frac{1}{1-ax} = \frac{1}{1-a} \Leftrightarrow 1-ax = 1-a \Leftrightarrow -ax = -a \Leftrightarrow -\frac{1}{a} = -a$$

$$\int_{-\infty}^0 \frac{x \cdot a}{(1-ax)^2} dx$$